

On the distribution of the zero sets of integer polynomials and beyond

My research interests for a general audience

Robert Wilms

University of Caen

Robert.wilms@unicaen.fr

April 10, 2024

Distribution of algebraic numbers

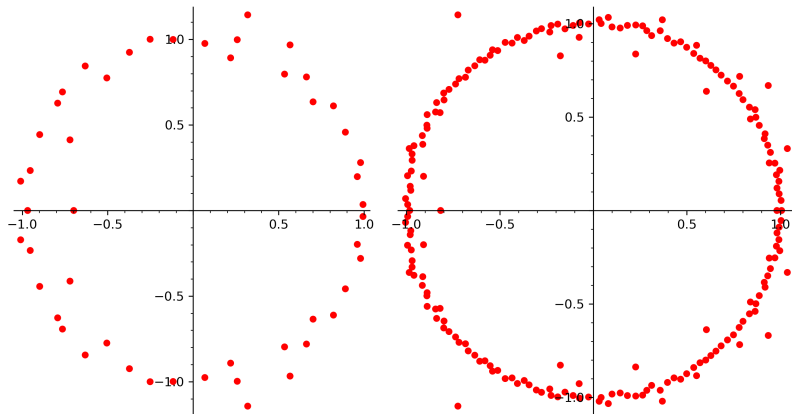
- Leopold Kronecker: *“Die ganzen Zahlen hat der liebe Gott gemacht, alles andere ist Menschenwerk.”*
(God made the integers, all else is the work of man.)
- Integers: $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$.
- Rational numbers: $\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}$.
- Real numbers $\mathbb{R}$: Limits of (Cauchy) sequences of rational numbers.
- Complex numbers: $\mathbb{C} = \mathbb{R} + i \cdot \mathbb{R}$ by adding $i = \sqrt{-1}$.
- Algebraic numbers: All numbers constructible in an algebraic way

$$\overline{\mathbb{Q}} = \left\{ z \in \mathbb{C} \mid \exists P = \sum_{k=0}^n a_k X^k \in \mathbb{Z}[X] \setminus \{0\} : P(z) = 0 \right\}.$$

- How to study the distribution of $\overline{\mathbb{Q}}$ in $\mathbb{C}$?
- For any sequence $(P_n = \sum_{k=0}^n a_{n,k} X^k)_{n \geq 1}$ with $|a_{n,k}| \leq M$ and $a_{n,n} \cdot a_{n,0} \neq 0$ the zero sets converge to the unit circle $S^1 \subseteq \mathbb{C}$.
- By Erdős–Turán’s Theorem, M can be replaced by functions $M(n, k)$ subexponential in n .

The zero set of a random polynomial I

- The zero sets of two polynomials with random integral coefficients in $\{-100, -99, \dots, 99, 100\}$ of degree 50 and 200 plotted with SageMath:



- What happens if the bounds $M(n, k)$ are not subexponential in n ?
- Can we get distribution with respect to the whole complex plane?

Zero sets distributed by the Fubini–Study measure

- For every $r \in \mathbb{R}$ and every positive integer n define the set

$$\mathcal{P}_{n,r} = \left\{ P = \sum_{k=0}^n a_k X^k \in \mathbb{Z}[X] \mid |a_k| \leq r \cdot \sqrt{\binom{n}{k}} \right\}.$$

Theorem 1 (W. 2022)

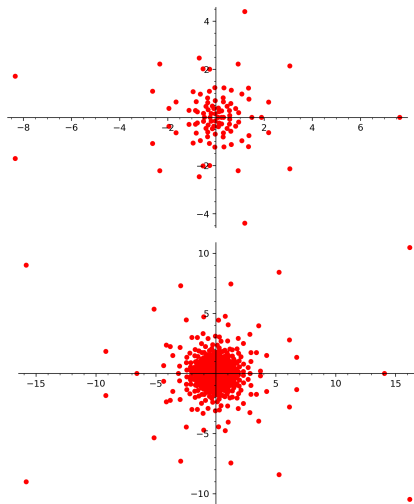
For every sequence $(r_n)_{n \geq 0}$ of real numbers with $\liminf_{n \rightarrow \infty} r_n^{1/n} > 1$ and every continuous function $f: \mathbb{C} \rightarrow \mathbb{C}$ with $\lim_{|z| \rightarrow \infty} f(z)$ being finite and well-defined,

$$\lim_{n \rightarrow \infty} \frac{1}{\#\mathcal{P}_{n,r_n}} \sum_{P \in \mathcal{P}_{n,r_n} \setminus \{0\}} \left| \frac{1}{n} \sum_{\substack{z \in \mathbb{C} \\ P(z)=0}} f(z) - \frac{i}{2\pi} \int_{\mathbb{C}} f(z) \frac{dz d\bar{z}}{(1 + |z|^2)^2} \right| = 0.$$

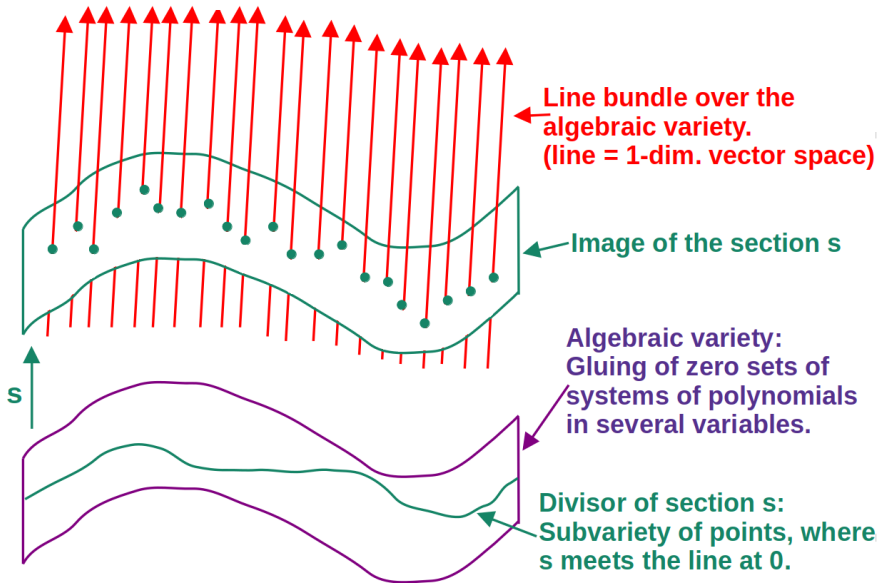
- That means, for “almost all” sequences $(P_n)_{n \geq 1}$ with $P_n \in \mathcal{P}_{n,r_n}$ the zero sets converge to the Fubini–Study measure.
- Note that $\sqrt{\binom{2m}{m}} \sim \frac{2^m}{(m\pi)^{1/4}}$ is exponential in $n = 2m$.

The zero set of a random polynomial II

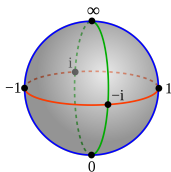
- The zero sets of two polynomials with random integral coefficients $-100\sqrt{\binom{n}{k}} \leq a_k \leq 100\sqrt{\binom{n}{k}}$ of degree $n = 100,700$ by SageMath:



Line bundles on algebraic varieties



Example: The projective line $\mathbb{P}_{\mathbb{C}}^1$



Adding a point at infinity, the complex numbers $\mathbb{C}$ become the Riemann sphere $\mathbb{P}_{\mathbb{C}}^1$ with homogeneous coordinates $[Z_0 : Z_1]$.

(Source of image: Wikipedia)

- We have a canonical isomorphism $\{Z_1 \neq 0\} \xrightarrow{\cong} \mathbb{C}, \quad [Z_0 : Z_1] \mapsto \frac{Z_0}{Z_1}.$
- $\mathbb{P}_{\mathbb{C}}^1$ is equipped with natural line bundles $\mathcal{O}(n)$ having global sections

$$H^0(\mathbb{P}_{\mathbb{C}}^1, \mathcal{O}(n)) = \left\{ \sum_{k=0}^n a_k Z_0^k Z_1^{n-k} \mid a_k \in \mathbb{C} \right\} \cong \left\{ P = \sum_{k=0}^n a_k X^k \in \mathbb{C}[X] \right\}$$

- The above identification respects the zero sets under $\{Z_1 \neq 0\} \xrightarrow{\cong} \mathbb{C},$

$$\operatorname{div} \left(\sum_{k=0}^n a_k Z_0^k Z_1^{n-k} \right) \cap \{Z_1 \neq 0\} \cong \left\{ z \in \mathbb{C} \mid P(z) = \sum_{k=0}^n a_k z^k = 0 \right\}.$$

Equidistribution of divisors of small sections

- $\mathcal{X}/\mathbb{Z}$: Algebraic variety over $\mathbb{Z}$ satisfying certain conditions of finiteness, compactness, separateness, flatness and smoothness.
- $\mathcal{L}$: Line bundle on $\mathcal{X}$ satisfying a certain condition of positivity (ample).
- h : Family of hermitian metrics h_x on every line $\mathcal{L}_x$ for every complex point $x \in \mathcal{X}(\mathbb{C})$ such that for every local section s the norm $|s(x)|^2 = h_x(s_x, s_x)$ varies smoothly in x and $c_1(\mathcal{L}, h) = \frac{\partial\bar{\partial}}{2\pi i} \log |s(x)|^2$ is a positive $(1, 1)$ form.
- Set $\bar{\mathcal{L}} = (\mathcal{L}, h)$ and $\hat{H}^0(\mathcal{X}, \bar{\mathcal{L}}) = \left\{ s \in H^0(\mathcal{X}, \mathcal{L}) \mid \sup_{x \in \mathcal{X}(\mathbb{C})} |s(x)| < 1 \right\}$.
Assume that $\hat{H}^0(\mathcal{X}, \bar{\mathcal{L}}^{\otimes n})$ generates the group $H^0(\mathcal{X}, \mathcal{L}^{\otimes n})$ for $n \gg 0$.

Theorem 2 (W. 2022)

If $d = \dim \mathcal{X}(\mathbb{C}) \geq 1$, then for every continuous $(d-1, d-1)$ form Φ on $\mathcal{X}(\mathbb{C})$

$$\lim_{n \rightarrow \infty} \frac{1}{\#\hat{H}^0(\mathcal{X}, \bar{\mathcal{L}}^{\otimes n})} \sum_{s \in \hat{H}^0(\mathcal{X}, \bar{\mathcal{L}}^{\otimes n}) \setminus \{0\}} \left| \frac{1}{n} \int_{\text{div}(s)(\mathbb{C})} \Phi - \int_{\mathcal{X}(\mathbb{C})} \Phi \wedge c_1(\mathcal{L}, h) \right| = 0.$$

Outlook on future research

- For every $x \in \overline{\mathbb{Q}}^d$ there exist d polynomials $P_1, \dots, P_d \in \mathbb{Z}[X_1, \dots, X_d]$, whose common zero locus is a finite set $S_x \subseteq \mathbb{C}^d$ containing x . We call the minimal such set S_x the Galois orbit $O(x)$ of x .
- Planned generalization of Theorem 1: For an explicit function $h: \overline{\mathbb{Q}}^d \rightarrow \mathbb{R}$, the set $\mathcal{P}_{n,r}^d = \{x \in \overline{\mathbb{Q}}^d \mid \#O(x) \leq n, h(x) \leq r\}$, and any continuous function $f: \mathbb{C}^d \rightarrow \mathbb{C}^d$ which extends to $\mathbb{P}_{\mathbb{C}}^d$,

$$\lim_{n \rightarrow \infty} \frac{1}{\#\mathcal{P}_{n,r}^d} \sum_{x \in \mathcal{P}_{n,r}^d} \left| \frac{1}{n} \sum_{z \in O(x)} f(z) - \int_{\mathbb{C}^d} f(z) \mu^d \right| = 0,$$

where $\mu = \frac{i}{2\pi} \left(\frac{\sum_{j=1}^d dz_j d\bar{z}_j}{1 + \sum_{j=1}^d |z_j|^2} - \frac{(\sum_{j=1}^d \bar{z}_j dz_j)(\sum_{j=1}^d z_j d\bar{z}_j)}{(1 + \sum_{j=1}^d |z_j|^2)^2} \right)$.

- Besides the standard norm $|\cdot|$, $\mathbb{Q}$ carries for every prime p the p -adic norm $|ap^n/b|_p = p^{-n}$, where $\gcd(a, p) = \gcd(b, p) = 1$. This induces a p -adic analysis on the algebraic closed and $|\cdot|_p$ -complete field $\mathbb{C}_p$.
- Planned work: Analogue of Theorem 2 for the p -adic analytic space $\mathcal{X}(\mathbb{C}_p)$ instead of $\mathcal{X}(\mathbb{C})$.

Thank you for your attention!